Proposed Research Directions

An Empirical Testing Framework for the Anomalous Presence Perception Hypothesis

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This document proposes a structured empirical testing framework for the Anomalous Presence Perception Hypothesis (Kapadia, 2025–2026; DOI: 10.5281/zenodo.19161509). It is submitted as a companion appendix to the hypothesis, outlining specific methodological approaches across five research categories. The proposals range from low-cost surveys requiring no laboratory access to neuroimaging studies requiring institutional resources. The author actively seeks collaboration with researchers who may wish to test any component of this framework — independently or jointly.

1. Surveys — Retrospective, Anonymous, Low-Risk

Survey-based approaches offer the lowest barrier to entry and can generate large, cross-cultural datasets without laboratory access or participant risk. The following surveys are proposed:

1.1 Bereavement Presence Survey

Target population: widows, widowers, and bereaved individuals who have lost a close family member. Core questions: Did you ever sense the deceased's presence after their death? How frequently? What triggered the experience — smell, sound, location, or spontaneous? How long after the death did experiences occur? Predicted outcome consistent with the hypothesis: experiences should peak in the first one to two years post-loss, triggered predominantly by olfactory and environmental cues associated with the deceased, consistent with the Neural Persistence and olfactory bypass models.

1.2 Haunted Location Survey

Collect standardised reports from visitors to locations culturally designated as haunted — schools, old homes, graveyards, historical sites. The hypothesis predicts that frequency of anomalous experience should correlate more strongly with sustained habitation by a specific individual than with the volume or recency of deaths at the location. This directly tests the Neural Persistence mechanism against a purely thermodynamic alternative.

1.3 Infrasound Exposure Survey

Target population: residents living within proximity of chronic infrasound sources — wind turbines, industrial machinery, subwoofers. Ask whether they have experienced unexplained unease, peripheral visual disturbances, or sensations of an unseen presence. Compare with a matched control group without infrasound exposure. Predicted outcome: infrasound-adjacent populations should report significantly higher rates of presence-type experiences.

1.4 Cultural Priming Survey

Compare paranormal experience rates across cultures with different ghost narratives — for example, India versus Sweden, or Japan versus the United Kingdom. The hypothesis predicts that cultures with richer, more specific ghost narratives will produce more defined presence experiences, while cultures with minimal ghost narrative will produce more ambiguous unease experiences from the same physical stimuli. This tests the conscious-subconscious integration model and the role of predictive processing in shaping the content of subconscious detections.

2. Existing Datasets — Secondary Analysis

Several existing datasets can be re-analysed to test specific predictions of the hypothesis without recruiting new participants.

2.1 Rees 1971 Dataset

The original Rees (1971) bereavement hallucination study of 293 widows and widowers in Wales represents the most directly relevant existing dataset. Re-analysis should focus on correlations between experience frequency and: age of bereaved individual, time elapsed since loss, depth of relationship closeness, and type of trigger reported. If accessible through the British Medical Journal archives, this dataset could directly test the Neural Persistence decay curve predicted by the hypothesis.

2.2 Longitudinal Grief Studies

Large longitudinal grief datasets from institutions including Harvard and Brown University frequently include questions about sensed presence as part of broader bereavement assessments. Re-analysis of these datasets with specific attention to presence experience frequency, timing, and triggers could provide a large-scale test of the hypothesis predictions without new data collection.

2.3 Infrasound and Health Databases

Public health records from communities near wind farms and industrial infrasound sources include self-reported anxiety, sleep disturbance, and unease data. Re-examining these databases specifically for presence-type experiences and correlating them with measured infrasound levels and exposure duration would provide direct epidemiological evidence for or against the infrasound-cortical arousal mechanism.

2.4 Temporal Lobe Epilepsy Registries

TLE patient registries systematically document sensed presence as an aura symptom. Comparing the frequency and characteristics of presence experiences in TLE patients against the hypothesis's Neural Persistence model — specifically whether presence content reflects the patient's most emotionally significant relationships — would provide neurological validation of the temporal lobe activation mechanism.

2.5 Heart Rate Variability Public Data

Existing HRV recordings from bereaved individuals in published stress research can be re-examined for patterns consistent with Cardiac Neural Persistence — specifically whether the autonomic nervous system shows residual synchronisation signatures with deceased partners in the months following loss.

3. Laboratory Studies — Healthy Volunteers

The following studies require laboratory access but do not require bereaved participants, minimising ethical complexity.

3.1 Infrasound Chamber Study

Expose healthy volunteers to infrasound at 10-18 Hz — below hearing threshold — in a controlled acoustic chamber. Measure self-reported presence sensations, skin conductance responses, heart rate variability, and EEG temporal lobe activity. Compare against a sham condition in which participants believe they are receiving infrasound but the chamber is silent. This directly tests the infrasound-cortical arousal mechanism while controlling for suggestion effects — addressing the primary limitation of Persinger's God Helmet studies.

3.2 EMF Exposure Study

Replicate and extend Persinger's God Helmet paradigm using weak magnetic fields targeted at the temporal lobes under rigorously double-blinded conditions. Unlike Persinger's original studies — which were criticised for insufficient blinding — this replication should follow the Granqvist et al. (2005) protocol while using Persinger's field duration and pattern parameters to resolve the existing replication dispute.

3.3 Olfactory DMN Activation

Present healthy volunteers with scents strongly associated with a specific living person — a partner's perfume, a parent's characteristic smell. Measure Default Mode Network activation via fMRI during olfactory presentation versus control scents. This tests the olfactory bypass mechanism — the hypothesis that smell activates the DMN's social simulation network more directly than any other sensory modality.

3.4 Predictive Processing VR Paradigm

Create a virtual environment in which a virtual person reliably appears at a specific location over repeated trials. Then remove the virtual person without warning. Measure how long the brain continues to expect their presence through reaction time, eye gaze patterns, and EEG anticipatory signals. This tests the predictive processing component of Neural Persistence in a controlled, non-grief context.

3.5 Cardiac Conditioning Baseline

In healthy couples, measure HRV synchronisation patterns when partners are physically together versus separated. Establish baseline signatures of cardiac co-regulation as a living reference point for Cardiac Neural Persistence research. This enables future comparison with bereaved individuals' HRV patterns in the absence of their partners.

4. Neuroimaging Studies — Bereaved Volunteers

These studies require ethics committee approval and bereaved volunteers, but represent the highest-value direct tests of the hypothesis.

4.1 fMRI of DMN During Bereavement

Show bereaved individuals photographs of the deceased versus photographs of strangers of similar age and appearance. Compare DMN activation — specifically the precuneus, posterior cingulate cortex, and medial prefrontal cortex. The hypothesis predicts significantly greater DMN activation for deceased individuals, reflecting the continued firing of Neural Persistence pathways even during passive visual presentation.

4.2 EEG During Spontaneous Presence Episodes

Recruit participants who report frequent spontaneous sensed presence experiences. Fit with mobile EEG and instruct to press a button when a presence is felt in their natural home environment. Examine EEG in the 10 seconds preceding the button press for temporal lobe activity consistent with the hypothesis. This is the most ecologically valid neuroimaging approach and may be achievable with consumer-grade mobile EEG devices.

4.3 MEG During Infrasound Exposure

Use magnetoencephalography during controlled infrasound exposure to precisely localise the brain regions generating presence-related activity. MEG provides millisecond temporal resolution and millimetre spatial precision — sufficient to distinguish temporal lobe activation from prefrontal or parietal responses — directly testing whether infrasound produces specifically temporal lobe responses consistent with the hypothesis mechanism.

5. Field Studies — Naturalistic, Non-Invasive

5.1 Infrasound Mapping of Reported Haunted Locations

Deploy calibrated low-frequency microphones in reported haunted locations and matched control buildings of similar age and construction. Correlate measured infrasound levels with frequency of reported experiences. The hypothesis predicts haunted locations will show elevated infrasound consistent with their structural features — large resonant spaces, aged ventilation systems, proximity to underground water.

5.2 HRV Field Study in Bereaved Individuals

Bereaved volunteers wear a chest-strap HRV monitor continuously for one week. They are instructed to press a dedicated button whenever they experience a sensed presence. Compare HRV patterns in the 10 seconds before the button press against baseline HRV throughout the week. The hypothesis predicts HRV anomalies preceding presence experiences, reflecting autonomic arousal from Cardiac Neural Persistence pathway activation.

5.3 Olfactory Residue Study

Collect air samples from recently vacated spaces — specifically the bedroom of a recently deceased individual. Present these air samples to healthy volunteers who never knew the deceased. Measure whether olfactory stimulation from residual volatile organic compounds produces detectable unease or presence-type responses, even in the absence of any personal neural map of the individual. This tests the purely thermodynamic component of the hypothesis independent of Neural Persistence.

6. Computational and Theoretical Modelling

These approaches require no human subjects and can be implemented with computational resources.

6.1 Predictive Processing Simulation

Build a Bayesian predictive processing model representing a brain with a strong prior expectation for a specific person's sensory signals. Remove the input stream and examine whether the model generates phantom predictions consistent with a sensed presence. This computational approach could directly replicate the Neural Persistence mechanism in silico and generate quantitative predictions about decay rate and trigger sensitivity.

6.2 Neural Network Persistence Model

Train an LSTM recurrent neural network on sequences representing a person's typical daily interaction patterns — timing, location, sensory signature. Remove the input stream entirely and measure how long the network continues to generate predictive outputs before settling to a neutral state. Compare the decay curve with the Rees (1971) data showing peak experiences in the first one to two years post-loss.

6.3 Thermodynamic Dispersion Model

Using standard diffusion equations, calculate how long the measurable thermal, chemical, and electromagnetic signatures of a human body's presence remain detectable in an enclosed living space following death. Test whether the calculated decay window aligns with the empirically observed one to two year peak in bereavement presence experiences. If the thermodynamic signature persists for approximately the same duration as the peak in reported experiences, this provides indirect support for the environmental energy component of the hypothesis.

Note on Collaboration

I developed this framework without institutional affiliation, laboratory access, or formal research training. I recognise that several proposed studies — particularly the neuroimaging and laboratory components — require resources, ethics approval, and expertise that I do not currently have access to. I am actively open to contact from researchers whose existing work or laboratory infrastructure aligns with any component of this framework — whether for independent replication, methodological critique, or direct collaboration. I am not asking to be handed a result. I am asking whether the questions are worth asking.

Author's Note: These research directions were developed independently — as a companion to the Anomalous Presence Perception Hypothesis — by a 17-year-old student from Ahmedabad, India. They emerged from the same process as the hypothesis itself: curiosity first, method later. I am not a trained researcher. I do not have laboratory access. What I have is a framework that I believe is worth testing, and the honesty to acknowledge that testing it requires people with more resources than I *currently have. These proposals are not a finished protocol. They are a starting point — submitted in the hope that someone finds the questions worth pursuing.* *ORCID: 0009-0002-1330-8664*

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